

SHL Assessment Recommendation System

Technical Solution Report

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1 Problem Statement

Develop an Gen AI recommendation system matching job requirements with SHL assessments. Requirements: Natural language processing(NLP), semantic search,API development, public accessibility, and deployment.

2 Technical Architecture

2.1 Core Components

Semantic Search: FAISS (Facebook AI Similarity Search) with Sentence Transformers (all-MiniLM-L6-v2) for embedding generation and semantic search. In-memory indexing of 348 assessments (584KB) extracted from the web.

API Layer: FastAPI 0.104.1 with Uvicorn ASGI server. Pydantic validation and auto-generated documentation at /docs.

Public Access: Cloudflare Tunnel for free HTTPS public URL without authentication.

2.2 System Flow

Query → Embedding Generation → FAISS Cosine Similarity Search → Ranking (0.0-1.0 scores) → Filtering (job.level, industry, skills) → JSON Response

3 Implementation

3.1 Search Algorithm

```
def search(query, max_results=10, filters=None):
    query_embedding = model.encode([query])
    distances, indices = index.search(query_embedding, k=max_results)
    similarities = 1 / (1 + distances)
    return filter_and_rank(indices, similarities, filters)
```

3.2 API Endpoints

GET /health	Health check
POST /recommend	Main recommendation
GET /docs	Interactive API docs

3.3 Request/Response Example

```
REQUEST: {"query": "Senior Java developer", "job_level": "Senior", "max_results": 5}
RESPONSE: {"total_results": 5, "recommendations": [
  {"assessment_name": "Java Programming Advanced", "relevance_score": 0.89}]}
```

4 Deployment Strategy

4.1 Evolution

Failed Attempts: Google Cloud Run (dependency conflict), Vertex AI Agent Engine (complexity), Cost: \$20-50/month.

Final Solution: Local FastAPI + Cloudflare Tunnel.

URL: <https://bishop-monitored-conf-heights.trycloudflare.com>. Setup: 2 minutes. Cost: \$0.

4.2 Deployment Scripts

`server.py` (local), `start_public.py` (ngrok), `start_cloudflare.py` (Cloudflare Tunnel).

5 Key Technical Decisions

FAISS: $O(\log n)$ search vs $O(n)$ brute force, memory efficient, no database needed.

Local Deployment: \$0 cost, 2-min setup, easy debugging, sufficient for demo.

Cloudflare Tunnel: No auth, free, HTTPS default, reliable CDN.

6 Results & Performance

Metrics: Search latency ≤ 100 ms, memory ≤ 500 MB, startup 5-7s, coverage 348/348 (100%).

Status: All endpoints operational, public URL active, Swagger UI accessible.

7 Challenges & Solutions

Challenge 1 - Dependency Conflict: `vertexai>=1.12.0` doesn't exist as standalone. *Solution:* Removed dependency, used local deployment.

Challenge 2 - Public URL: ngrok requires auth. *Solution:* Cloudflare Tunnel with zero-config.

Challenge 3 - Download Timeouts: Cloudflared binary (18MB) timing out. *Solution:* Non-blocking port check, local caching, pre-download via curl.

8 Conclusion

Successfully deployed production-ready system with semantic search (FAISS), RESTful API (FastAPI), public access (Cloudflare), zero cost, and 2-minute setup.

Deliverables:

- Public API: <https://bishop-monitored-conf-heights.trycloudflare.com>
- Documentation: Interactive Swagger UI at `/docs`
- Source: GitHub repository with deployment scripts

Future Enhancements: Redis caching, user feedback loop, permanent hosting (Render.com), A/B testing for embedding models, rate limiting/authentication.